

STRATEGIC WORKFORCE ATTRITION ANALYSIS

Project Summary Report

Prepared on completion of project activities

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1. Executive Summary

This report documents the work completed under the Strategic Workforce Attrition Analysis project. The project brought together five separate workforce data sources — Employee, Department, Manager, Salary, and Attrition records — and prepared them for analysis. Data quality issues such as duplicate records, invalid codes, inconsistent text formats, and missing values were identified and addressed across all five datasets.

Following data preparation, the cleaned data was analyzed to answer the business questions defined in the original project brief. The analysis covered workforce composition, compensation and performance patterns, attrition trends, exit interview sentiment, and departmental budget information. Twelve charts were produced to present the findings in a visual, easy-to-understand format.

The final employee population analyzed consisted of 1,200 unique employee records, of which 239 employees (about 20%) were recorded as having left the organization. The cleaned datasets and supporting analysis are available as the output of this project.

2. Project Overview

The purpose of this project was to examine employee attrition within the organization using existing workforce records. Prior to this project, employee, department, manager, salary, and exit-event data existed separately and had not been reviewed together. The project brief identified five core questions to be answered: the overall scale of attrition, which departments and roles are most affected, the main reasons employees leave, the relationship between exit sentiment and departure reason, and whether performance, overtime, or salary patterns relate to attrition.

This report summarizes the work carried out to prepare the data and answer these questions. It reflects only what was completed during the project, based on the analysis performed; it does not include predictions, external benchmarking, or recommendations, as these were outside the agreed scope of the project.

3. Objectives Addressed

The table below lists the eight objectives defined in the project brief and describes how each was addressed during the project.

#	Objective (from Project Brief)	How It Was Addressed
1	Consolidate and prepare workforce data from five separate sources into a unified, analysis-ready format.	Addressed. All five datasets (Employee, Department, Manager, Salary, Attrition) were reviewed, cleaned, and prepared individually, and combined where needed (e.g., employee records

#	Objective (from Project Brief)	How It Was Addressed
		linked to department records) to support the analysis.
2	Identify and resolve data quality issues, including duplicates, invalid identifiers, inconsistent formats, and missing entries.	Addressed. Duplicate records were found and removed in four of the five datasets. Invalid placeholder codes, inconsistent text formats (e.g., gender labels, salary values), and missing values were identified and corrected or flagged as missing.
3	Establish referential integrity across the employee, department, manager, salary, and attrition datasets.	Partially addressed. Employee IDs in the Attrition dataset and department IDs in the Manager dataset were checked against their respective reference tables and confirmed to match. Invalid department and manager codes found in the Employee dataset were identified and treated as missing values rather than reassigned to valid codes.
4	Quantify the overall scale and distribution of employee attrition across the organization.	Addressed. The overall attrition rate, along with the count of employees who stayed versus left, was calculated and visualized.
5	Examine attrition patterns across departments, job roles, salary levels, and time periods.	Mostly addressed. Attrition was examined by department, by job role, and by year. A direct comparison of attrition against salary level was not produced; average salary was instead examined by job role and department.
6	Analyze the reasons behind employee departures and their relationship to exit interview sentiment.	Addressed. The frequency of each departure reason was calculated, and average exit interview scores were compared across reasons.
7	Assess workforce characteristics — gender, performance ratings, overtime hours, and salary — in the context of attrition.	Partially addressed. Gender, performance rating, and salary distributions were summarized for the overall workforce. Overtime hours were cleaned but not separately analyzed. These characteristics were not directly cross-compared against attrition status.
8	Produce visual summaries of key workforce metrics for non-technical stakeholders.	Addressed. Twelve charts were produced covering gender, job roles, salary, attrition, departments, performance ratings, and departure reasons.

4. Dataset Overview

Five datasets were used in this project, each representing a different aspect of the workforce. The table below shows the size of each dataset before and after the data cleaning steps described in Section 6.

Dataset	Original Records	Records After Cleaning	Description
Employee	1,205	1,200	Personal details, hire date, salary, department and manager assignment, job role, overtime, performance rating, and attrition status.
Department	12	12	Department name, regional location, and budget allocation.
Manager	46	45	Manager name, assigned department, hire date, and performance rating.
Salary	2,404	2,403	Salary history records, including effective dates, monthly salary amounts, and reason for change.
Attrition	241	240	Exit events, including departure date, stated reason for leaving, and exit interview score.

In total, the analysis was based on 1,200 unique employees, 12 departments, 45 managers, 2,403 salary history records, and 240 recorded attrition (exit) events.

5. Analysis Performed

The analysis was organized around the twelve business questions set out in the project brief, grouped into five themes.

Workforce Composition

The total number of active employees was counted, and the workforce was broken down by gender, job role, and department.

Compensation & Performance

Average salary was calculated by job role and by department, and the spread of performance ratings across the workforce was examined.

Attrition Patterns

The proportion of employees who left versus those who stayed was calculated. Attrition was then broken down by job role and by department, and tracked across years to identify whether departures were increasing or decreasing over time.

Exit Analysis

The frequency of each stated reason for leaving was calculated, and exit interview scores were reviewed in aggregate and broken down by reason for leaving.

Departmental Budget

Department budgets were reviewed to identify the highest-funded department, the average budget across departments, and how departments are distributed across geographic regions.

6. Observations

The following data quality issues were observed and addressed while preparing the five datasets for analysis.

Employee Dataset

- 5 duplicate employee records were found and removed, reducing the dataset from 1,205 to 1,200 records.
- Gender values appeared in nine different inconsistent formats (e.g., "Male", "male", "M", "Man", "f"). These were standardized into two consistent categories: Male and Female.
- Salary values were recorded in several different formats, including plain numbers, abbreviated amounts (e.g., "85k"), values written out in words (e.g., "fifty two thousand"), and one entry recorded as "zero". All values were converted into a single consistent numeric format, and missing or invalid entries were filled using the median salary for the matching job role.
- Department and manager assignments contained invalid placeholder codes (department code "99" and manager code "9999"). These were identified and treated as missing values.
- Overtime hours contained two negative values, which were corrected to missing and then filled using the median value for the matching job role.
- Missing performance ratings were filled using the median rating for the matching job role.

Department Dataset

- No duplicate records or invalid budget values were found.
- A check for unusually high or low budget values found none in this dataset.

Manager Dataset

- 1 duplicate manager record was found and removed, reducing the dataset from 46 to 45 records.
- 1 missing performance rating was filled using the median rating across all managers.
- Department codes recorded against each manager were checked against the Department dataset and all were found to be valid.

Salary Dataset

- 1 duplicate salary record was found and removed, reducing the dataset from 2,404 to 2,403 records.
- 10 records had a missing effective date that could not be recovered from the available data.
- A check for unusually high or low salary values found none in this dataset.

Attrition Dataset

- 1 duplicate exit event record was found and removed, reducing the dataset from 241 to 240 records.
- All employee IDs in this dataset were checked against the Employee dataset and confirmed to be valid; one employee (ID 2102) was found to have two separate recorded departure events.
- One exit interview score was recorded as a negative value; this was corrected and, along with other missing scores, filled using the median score.

7. Key Findings and Insights

Workforce Composition

- The organization's active workforce, after cleaning, consists of 1,200 employees across 12 departments and 7 job roles.
- Gender representation across the cleaned workforce is relatively balanced between Male and Female employees.

Compensation & Performance

- Among departments, HR has the highest average salary (approximately \$97,725), while Product has the lowest (approximately \$81,706).
- Performance ratings are concentrated mainly in the mid-to-high range, with ratings of 3 and 4 being the most common.

Attrition Patterns

- Of the 1,200 employees analyzed, 239 (about 20%) are recorded as having left the organization, while 961 (about 80%) remained.
- Engineering recorded the highest number of departures of any department (42), followed by Sales (29).
- Recorded attrition events increased year over year, from 11 in 2018 to 56 in 2024, based on the 240 dated exit events.

Exit Analysis

- Retirement was the most frequently cited reason for leaving (46 cases), followed by a better offer elsewhere (43 cases), relocation (40 cases), and management issues (40 cases).
- The average exit interview score across all departures was approximately 3 out of 5, with a score of 3 being the single most common response (67 of 240 cases).
- Average sentiment varied by reason for leaving: employees who left for a career change reported the highest average score (3.25), while those who left for a better offer reported the lowest (2.63).

Departmental Budget

- IT Infrastructure holds the highest departmental budget (approximately \$4.86 million), followed by R&D (approximately \$4.13 million).
- The average departmental budget across all 12 departments is approximately \$2.47 million.
- Departments are concentrated mainly in the East region (5 departments) and Remote operations (4 departments), with fewer departments based in the West (2) and South (1).

8. Deliverables Achieved

The table below compares the deliverables defined in the original project brief against what was completed.

Expected Deliverable	Outcome
Cleaned Employee dataset	Achieved. Duplicates removed; gender and salary values standardized. Invalid department/manager codes were identified and recorded as missing values rather than reassigned.
Cleaned Department dataset	Achieved. No duplicates or invalid budget values were present.
Cleaned Manager dataset	Achieved. Duplicate removed; missing performance rating filled; department codes verified.
Cleaned Salary dataset	Achieved. Duplicate removed; no invalid or outlier salary values found.
Cleaned Attrition dataset	Achieved. Duplicate removed; reason values standardized; invalid exit score corrected.
Summary of workforce metrics (headcount, gender, job role, salary, performance)	Achieved, through the Workforce Composition and Compensation & Performance analysis.
Summary of departmental	Achieved, through the Departmental Budget analysis.

Expected Deliverable	Outcome
characteristics (employee counts, budget, region)	
Summary of attrition patterns (rate, trend, job role, department, reasons)	Achieved, through the Attrition Patterns and Exit Analysis sections.
Summary of exit interview sentiment	Achieved. Average scores were calculated overall and by departure reason.
Twelve visual charts covering key workforce metrics	Achieved. All charts listed in the project brief were produced, covering gender, job roles, salary, attrition status, departments, performance ratings, and departure reasons.

9. Conclusion

This project consolidated five separate workforce data sources, identified and resolved a range of data quality issues, and used the cleaned data to answer the twelve business questions set out in the project brief. The resulting analysis provides a consolidated view of the organization's workforce composition, compensation patterns, attrition trends, and exit sentiment, supported by twelve charts.

Of the eight project objectives, six were fully addressed and two were partially addressed: full referential integrity between the employee dataset and the department/manager reference tables was not separately verified, and a small number of workforce characteristics (overtime hours, and the cross-comparison of gender, performance, and salary against attrition status) were prepared but not directly analyzed against attrition.

In line with the original project scope, this report does not include predictive modeling, external benchmarking, individual employee assessments, or recommendations for policy or compensation changes, as these areas were explicitly excluded from the project brief.